

Simulation study design for investigating Random Forests for Variable Importance

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1 Purpose of simulation study

The purpose of the present simulation study is to investigate the performance of various variable importance measures for random forests with respect to identifying effects in a sparse setting.

2 Simulation setting

We will simulate from a logistic regression model with a single binary outcome, Y and 52 binary covariates, $(X_1, \dots, X_{52} = X)$.

$$EY = \text{logis}(\theta_0 + \theta X) \tag{1}$$

The simulation engine is created from the ECG + OHCA dataset using the `synthesize` function of the `riskRegression` package. For this reason, the correlation structure of the covariates and the effect sizes are similar to this case.

3 Variable importance measures

For random forests we will consider permutation-based variable importance as well as minimal depth and average treatment effect as variable importance measures. For a comparison outside the random forest framework, we will use average treatment effect within a logistic regression model.

4 Notions of scoring

Exactly how performance of variable importance measures is to be scored is unclear to me at this time. A non-parametric method of scoring would be preferable, as different notions of importance might not be directly comparable. If each variable importance measure admits a ranking (preferably allowing for ties), the accuracy of importance rankings may be evaluated against a ground

truth of importance rank. The task of selecting a subset of important variables from a large set of mostly unimportant variables may be different than ranking the importance of a relatively small set of variables that is known to have some importance. These two tasks may be in tension for some methods. Perhaps it should be further clarified what kind of task we wish to score with respect to.